

12 (**long**)

(a) A thin-walled cylindrical vessel of radius r , thickness t , and fracture toughness K_{IC} is pressurised with gas of internal pressure P . Derive an expression for the critical flaw size a_{crit} . What will happen if $a_{crit} > t$? [8]

(b) Silicon nitride rods with constant cross-sectional area $A = 0.25 \text{ m}^2$ are subjected to constant tensile stress. A rod of length 1 m fails at 0.9 GPa with a failure probability of 0.01. A rod of length 0.4 m fails at 1 GPa with the same failure probability. Using the Weibull statistics of fracture from the Materials data book, find the stress that will cause a rod of length 0.2 m to fail with a probability of 0.01. [10]

(c) A turbine blade made of the silicon nitride from part (b) is attached to a rotor of radius $a = 0.2 \text{ m}$, as shown in Figure 10. The axial stress distribution in the turbine blade can be approximated as:

$$\sigma(x) = -6x + 2.4 \text{ GPa}$$

where x is the distance from the centre of the rotor. The blade has a tip radius $b = 0.4 \text{ m}$ with constant cross-sectional area $A_b = 0.1 \text{ m}^2$. Use the results from part (b) to find the failure probability of the blade. [12]

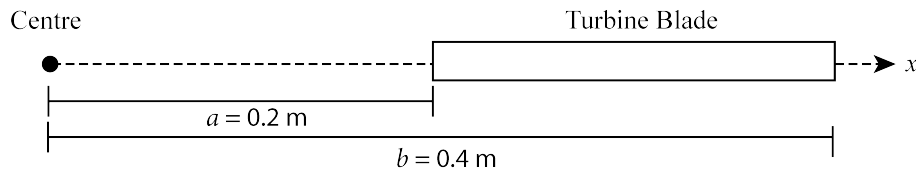


Fig. 10

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